1. Overview

This technical reference documents the exact material thermophysical properties, natural cooling parameters, radiation coefficients, forced coolant flow parameters, and volumetric heat source definitions utilized in the three-dimensional finite element model (COMSOL Multiphysics). It serves as the companion data sheet to the governing equations and mathematical formulations for publication in research papers.

2. Material Thermophysical Properties

The multi-domain assembly consists of five distinct material domains: the active wound metallized film core (Ω_{core}), encapsulating potting epoxy resin (Ω_{epoxy}), outer polyphenylene sulfide housing (Ω_{PPS}), the aluminum cooling box enclosure (Ω_{Al}) containing the overall oil cooling setup, and the circulating dielectric coolant oil (Ω_f).

2.1 Orthotropic Thermal Conduction in the Wound Film Core

The active core comprises bi-axially oriented metallized polypropylene (BOPP) film wound in tightly packed concentric layers. Because thermal conduction is facilitated in-plane by continuous metallic electrodes (aluminum/zinc spray and metallization) but impeded across layers by polymer-to-polymer thermal contact resistance, the core exhibits pronounced orthotropic thermal conductivity:

$$\mathbf{k}_{\text{core}} = \begin{bmatrix} k_{xx} & 0 & 0 \\ 0 & k_{yy} & 0 \\ 0 & 0 & k_{zz} \end{bmatrix} = \begin{bmatrix} 1.1362 & 0 & 0 \\ 0 & 0.2200 & 0 \\ 0 & 0 & 0.2200 \end{bmatrix} \text{W m}^{-1} \text{K}^{-1} \tag{1}$$

where $k_{xx} = 1.1362\,\text{W m}^{-1}\,\text{K}^{-1}$ represents longitudinal/in-plane heat conduction parallel to the winding foil, and $k_{yy} = k_{zz} = 0.2200\,\text{W m}^{-1}\,\text{K}^{-1}$ denotes transverse cross-layer heat conduction through stacked dielectric films.

2.2 Consolidated Material Properties Matrix

Table 1: Thermophysical Material Properties Assigned Across Simulation Domains.

Domain	**Material / Role**	**Density** ρ [kg m^{-3}]	**Specific Heat** C_p [J kg^{-1} K^{-1}]	**Thermal Conductivity** k [W m^{-1} K^{-1}]	**Viscosity** μ [Pa s]
Ω_{core}	Wound BOPP Film Core	907.17	1888.14	$k_{xx} = 1.1362,\ k_{yy,zz} = 0.2200$	— (Solid)
Ω_{epoxy}	Potting Epoxy Resin	1820.00	1340.00	$k_{\text{iso}} = 0.1500$	— (Solid)
Ω_{PPS}	Case Housing (PPS)	1650.00	1000.00	$k_{\text{iso}} = 15.0000$	— (Solid)
Ω_{Al}	Aluminum Cooling Enclosure Box	2700.00	900.00	$k_{\text{iso}} = 238.0000$	— (Solid)
Ω_f	Dielectric Coolant Oil	880.00	1900.00	$k_f = 0.1300$	0.0250 (25 cP)

3. Natural Cooling and Surface Radiation Parameters

On external case boundaries (Γ_{ext}) exposed to ambient surroundings, heat transfer occurs through combined natural convection and thermal radiation:

$$q_{\text{ext}} = h_{\text{nat}}(T - T_{\text{ext}}) + \varepsilon_{\text{rad}}\sigma_{\text{SB}}(T^4 - T_{\text{amb}}^4) \tag{2}$$

- **Natural Convection Heat Transfer Coefficient (h_{nat}):** Prescribed as $h_{\text{nat}} = 9.49\,\text{W m}^{-2}\,\text{K}^{-1}$, characterizing buoyancy-induced natural air convection over vertical and horizontal casing surfaces under standard indoor operating conditions.
- **Surface Emissivity (ε_{rad}):** Set to $\varepsilon_{\text{rad}} = 0.85$ on outer polymer casing surfaces, consistent with high-emissivity non-metallic industrial polymer enclosures in the long-wave infrared spectrum.

4. Internal Heat Generation and Electrical Operating Losses

Internal thermal dissipation arises from equivalent series resistance (ESR) Joule heating in metallized electrodes and dielectric polarization dissipation in the polypropylene polymer film:

$$P_{\text{loss}} = I_{\text{rms}}^2 \cdot \text{ESR} + 2\pi f C V_{\text{rms}}^2 \tan\delta \tag{3}$$

- **Nominal Rated Heat Dissipation ($P_{\mathbf{rated}}$):** $P_{\text{total}} = 0.714\,\text{W}$ for the full capacitor assembly. In the half-symmetry domain, the uniform volumetric heat source rate is:

$$P_0 = \int_{\Omega_{\text{core}}} \dot{q}_v \, dV = 0.357\,\text{W} \tag{4}$$